

TASK 1

Question: To execute large pipelines, launch multiple jobs, measure stability, monitor resources, detect failures, and analyze performance.

```
top - 07:23:11 up 1:23, 1 user, load average: 0.03, 0.04, 0.00
Tasks: 29 total, 1 running, 28 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.1 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3855.3 total, 3385.6 free, 469.0 used, 134.1 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used, 3386.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1976	ganesh	20	0	8188	6020	3964	R	0.7	0.2	0:00.15	top
1	root	20	0	24560	15472	11572	S	0.0	0.4	0:04.89	systemd
2	root	20	0	3760	2336	2204	S	0.0	0.1	0:00.09	init-systemd(Ub
3	root	20	0	3612	2096	2004	S	0.0	0.1	0:00.00	init-watcher
9	root	20	0	3760	2108	2004	S	0.0	0.1	0:00.00	init
70	root	19	-1	50652	17180	15868	S	0.0	0.4	0:01.57	systemd-journal
107	systemd+	20	0	22416	14452	11956	S	0.0	0.4	0:00.27	systemd-resolve
118	root	20	0	34708	11976	9220	S	0.0	0.3	0:03.03	systemd-udev
150	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.21	chronyd-starter
154	root	20	0	4472	3112	2868	S	0.0	0.1	0:00.06	cron
159	message+	20	0	8796	5452	4736	S	0.0	0.1	0:00.35	dbus-daemon
170	root	20	0	44096	29700	14552	S	0.0	0.8	0:01.05	networkd-dispat
177	root	20	0	18440	9404	8160	S	0.0	0.2	0:00.44	systemd-logind
196	syslog	20	0	220548	5556	4620	S	0.0	0.1	0:00.35	rsyslogd
247	_chrony	20	0	23588	11072	7092	S	0.0	0.3	0:00.41	chronyd
264	_chrony	20	0	12096	2464	1824	S	0.0	0.1	0:00.01	chronyd
283	root	20	0	123460	32652	18048	S	0.0	0.8	0:00.62	unattended-upgr
296	root	20	0	5492	2764	2576	S	0.0	0.1	0:00.03	agetty
330	root	20	0	3764	1304	1176	S	0.0	0.0	0:00.00	SessionLeader
331	root	20	0	3832	1436	1240	S	0.0	0.0	0:00.25	Relay(332)
332	ganesh	20	0	6528	5848	3864	S	0.0	0.1	0:00.94	bash
333	root	20	0	8648	5516	4672	S	0.0	0.1	0:00.03	login
399	ganesh	20	0	22348	13564	10932	S	0.0	0.3	0:00.57	systemd

```
ganesh@DESKTOP-79N3772:~/module_test$ kill %1 %2 %3
-bash: kill: %1: no such job
[2]- Terminated sleep 120
[3]+ Terminated sleep 150
ganesh@DESKTOP-79N3772:~/module_test$
```

Observation

Pipelines and multiple background jobs were executed, system resources were monitored, failures were detected, and execution performance was measured using time.

TASK 2

Question: To document architecture, create flow diagrams, explain execution paths, describe syscalls, record design decisions, and review documentation.

```
ganesh@DESKTOP-79N3772: ~/module_test
ganesh@DESKTOP-79N3772:~/module_test$ nano flow.txt
ganesh@DESKTOP-79N3772:~/module_test$ cat architecture.txt
SYSTEM ARCHITECTURE

User
|
v
Shell
|
v
Command
|
v
Process
|
v
Kernel
|
v
System Resources

Execution Path:
User -> Shell -> Process -> Kernel -> Resource

System Calls:
fork() - Create process
exec() - Execute program
wait() - Wait for process
exit() - Terminate process
ganesh@DESKTOP-79N3772:~/module_test$
```

```

User -> Shell -> Process -> Kernel -> Resource

System Calls:
fork() - Create process
exec() - Execute program
wait() - Wait for process
exit() - Terminate process
ganesh@DESKTOP-79N3772:~/module_test$ cat flow.txt
START
|
v
Receive Command
|
v
Create Process
|
v
Execute Command
|
v
Check Result
/      \
Success  Failure
|        |
v        v
END      Error Handling
          |
          v
          END
ganesh@DESKTOP-79N3772:~/module_test$

```

```

ganesh@DESKTOP-79N3772:~/module_test$ nano decisions.txt
ganesh@DESKTOP-79N3772:~/module_test$ ls -l
total 60
-rw-r--r-- 1 ganesh ganesh 297 Oct 5 07:25 architecture.txt
-rw-r--r-- 1 ganesh ganesh 261 Oct 5 07:27 decisions.txt
-rwxr-xr-x 1 ganesh ganesh 192 Oct 5 06:01 error_test.sh
-rw-r--r-- 1 ganesh ganesh 223 Oct 5 07:25 flow.txt
-rw-r--r-- 1 ganesh ganesh 0 Oct 5 06:02 input.txt
-rwxr-xr-x 1 ganesh ganesh 50 Oct 5 05:43 module1.sh
-rwxr-xr-x 1 ganesh ganesh 57 Oct 5 05:43 module2.sh
-rwxr-xr-x 1 ganesh ganesh 17312 Oct 5 07:13 test
-rw-r--r-- 1 ganesh ganesh 116 Oct 5 07:13 test.c
-rwxr-xr-x 1 ganesh ganesh 227 Oct 5 07:16 test_cases.sh
-rw-r--r-- 1 ganesh ganesh 56 Oct 5 07:17 test_report.txt
-rw-r--r-- 1 ganesh ganesh 212 Oct 5 07:13 valgrind_report.txt
ganesh@DESKTOP-79N3772:~/module_test$

```

Observation

The system architecture, execution flow, system calls, and design decisions were documented and reviewed using text files.

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TASK 1

Question: To organize source code, improve build scripts, add README, verify repository structure, tag releases, and validate deliverables.

```
ganesh@DESKTOP-79N3772:~/module_test$ mkdir project
ganesh@DESKTOP-79N3772:~/module_test$ cd project
ganesh@DESKTOP-79N3772:~/module_test/project$ mkdir src
ganesh@DESKTOP-79N3772:~/module_test/project$ nano src/main.sh
ganesh@DESKTOP-79N3772:~/module_test/project$ chmod +x src/main.sh
ganesh@DESKTOP-79N3772:~/module_test/project$ nano README.md
ganesh@DESKTOP-79N3772:~/module_test/project$ nano build.sh
ganesh@DESKTOP-79N3772:~/module_test/project$ chmod +x build.sh
ganesh@DESKTOP-79N3772:~/module_test/project$ ./build.sh
Building project...
Build completed successfully
ganesh@DESKTOP-79N3772:~/module_test/project$ find . -maxdepth 2 -type f
./src/main.sh
./README.md
./build.sh
ganesh@DESKTOP-79N3772:~/module_test/project$
```

Observation: Source code, build scripts, README, and repository structure were organized and a release tag was created.

TASK 2

Question: To create demo scenarios, validate features, rehearse presentation, demonstrate pipelines, demonstrate signals, and perform final project review.

```
ganesh@DESKTOP-79N3772:~/module_test/project$ ./build.sh
Building project...
Build completed successfully
ganesh@DESKTOP-79N3772:~/module_test/project$ find . -maxdepth 2 -type f
./src/main.sh
./README.md
./build.sh
ganesh@DESKTOP-79N3772:~/module_test/project$ ./build.sh
Building project...
Build completed successfully
ganesh@DESKTOP-79N3772:~/module_test/project$ ./src/main.sh
Project executed successfully
ganesh@DESKTOP-79N3772:~/module_test/project$ echo -e "apple\nbanana\napple\norange" | sort | uniq -c
  2 apple
  1 banana
  1 orange
ganesh@DESKTOP-79N3772:~/module_test/project$ sleep 100 &
[1] 2201
ganesh@DESKTOP-79N3772:~/module_test/project$ kill -STOP %1

[1]+  Stopped                  sleep 100
ganesh@DESKTOP-79N3772:~/module_test/project$ jobs
[1]+  Stopped                  sleep 100
ganesh@DESKTOP-79N3772:~/module_test/project$ kill %1

[1]+  Stopped                  sleep 100
[1]+  Terminated              sleep 100
ganesh@DESKTOP-79N3772:~/module_test/project$
```

Observation: Demo scenarios, pipelines, background processes, and signals were tested successfully, followed by final project verification.

